RIVER VALLEY HIGH SCHOOL
YEAR 6 PRELIMINARY EXAMINATION II

CANDIDATE NAME					
CENTRE NUMBER		INDEX NUMBER		CLASS	

H2 BIOLOGY **9648/02**

Paper 2 Core Paper **16 September 2015**

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams and graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Section A
Answer **all** questions in the spaces provided on the question paper.

Section B
Answer any **one** question on the answer paper provided.
Circle the question attempted on the cover page.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together. The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	**/ 13**
2	**/ 9**
3	**/ 11**
4	**/ 8**
5	**/ 9**
6	**/ 10**
7	**/ 10**
8	**/ 10**
Section B	
9 or 10*	**/ 20**
Total	

This Question Paper consists of **20** printed pages.

Section A (80 marks)

Answer **all** the questions in this section.

1. Fig 1.1 is an electron microgram showing part of the cell body of a neurone.

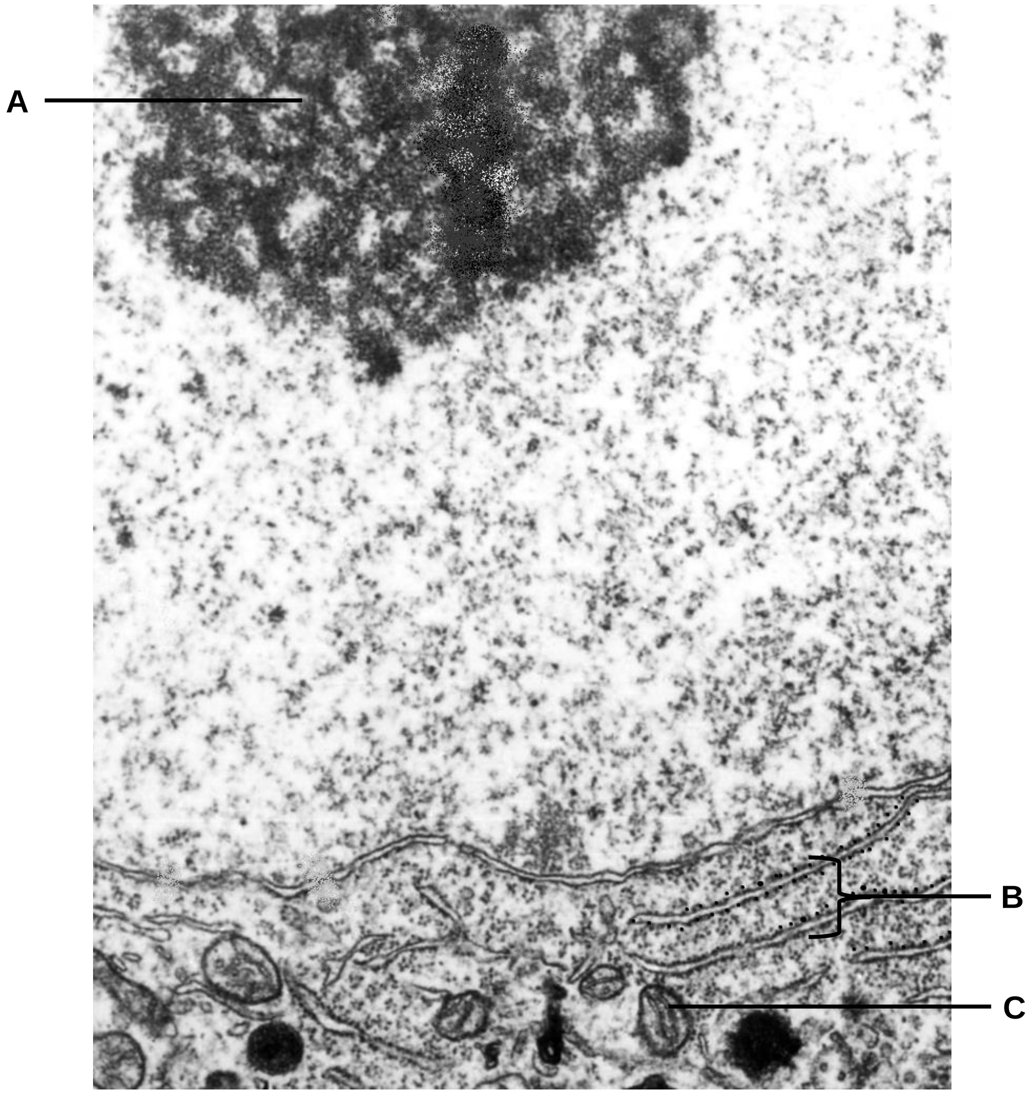

Fig 1.1

(adapted from http://www.columbia.edu/itc/hs/medical/sbpm_histology_old/lab/micro_popup15.html)

(a) Name the structures labelled **A, B** and **C**. [3]

A ______________________________

B ______________________________

C ______________________________

Dopamine is a neurotransmitter associated with the ‘pleasure system’ of the brain, providing feelings of enjoyment and reinforcement to motivate us to do, or continue doing certain activities.

Dopamine is synthesised in dopaminergic neurones via a multi-step process involving enzymes tyrosine hydroxylase and aromatic amino acid decarboxylase (AAAD).

An outline of the biochemical pathway for the synthesis of dopamine is shown in Fig 1.2.

$$\text{Tyrosine} \xrightarrow{\text{Tyrosine hydroxylase}} \text{L-DOPA} \xrightarrow{\text{AAAD}} \text{Dopamine}$$

Fig 1.2

(b) With reference to Fig 1.1 and 1.2, suggest the role(s) of structures **A**, **B** and **C** in the synthesis of dopamine. [4]

Fig 1.3 shows a drawing of a synapse between two dopaminergic neurones.

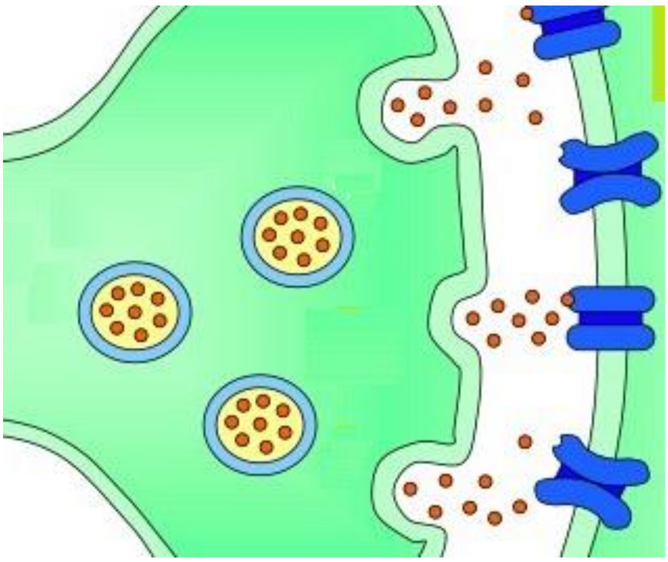

Fig 1.3

(c) Describe the processes shown in Fig 1.3**,** which leads to the depolarisation of the post-synaptic membrane. [4]

Excessive neurotransmission of dopamine is associated with schizophrenia, a clinical condition marked by disordered thoughts.

Neuroleptics are a class of compounds with a high affinity for dopamine receptors and are used to alleviate symptoms of the condition.

(d) Suggest how neuroleptics work to alleviate symptoms of schizophrenia. [2]

[Total: 13]

2. Fig. 2.1 is a photomicrograph of a root tip of *Allium cepa*.

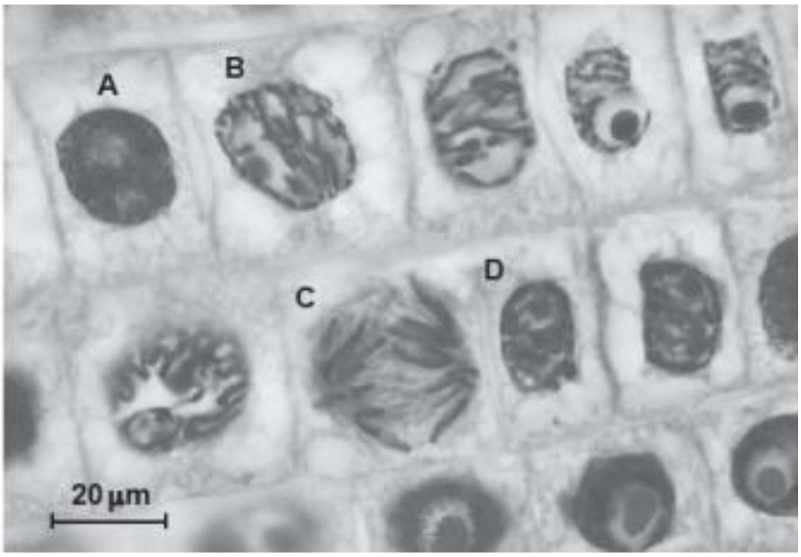

Fig. 2.1

(a) Describe the events that occur during stage **B**. [2]

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(b) Explain the significance of stage **C** in a growing plant root tip. [3]

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(c) Distinguish between the terms '*sister chromatids'* and '*homologous chromosomes'.* [2]

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In normal cells, the cell cycle is controlled at checkpoints. Fig. 2.2 shows the main stages in the cell cycle and the three checkpoints.

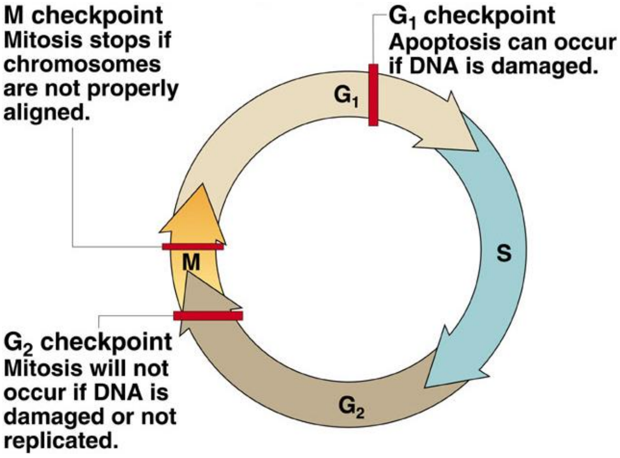

Fig. 2.2

(d) With reference to Fig. 2.2, suggest the consequences of dysregulation of **M** checkpoint. [2]

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[Total: 9]

3. Fig. 3.1 is an electron micrograph showing the process of protein synthesis in a prokaryote.

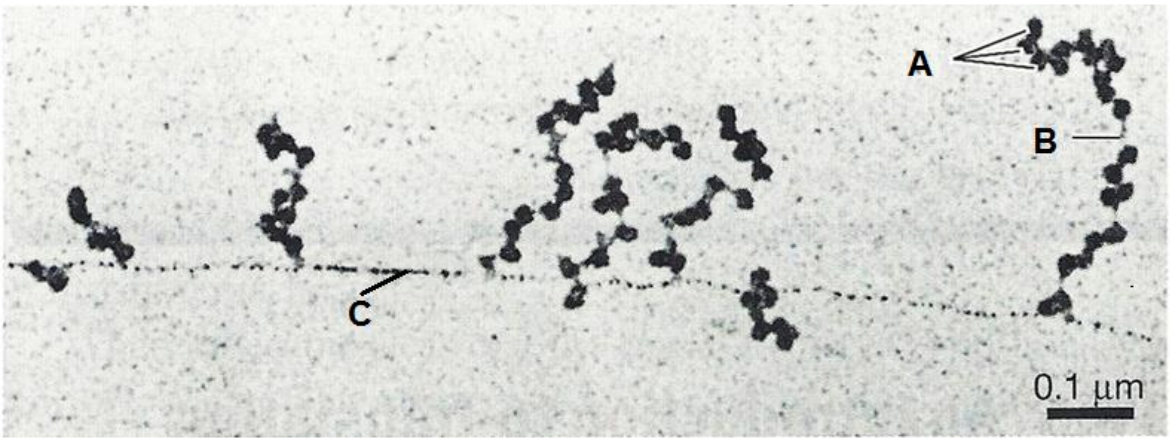

Fig. 3.1

(a) Identify structures **A** and **C**. [2]

A ______________________________

C ______________________________

(b) Describe how structure **B** is synthesised in prokaryotes. [3]

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(c) Describe how structure **A** is adapted to its function in protein synthesis. [2]

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(d) With reference to Fig 3.1, describe and explain how the pattern of protein synthesis is different from that in eukaryotes. [2]

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(e) Comparing to prokaryotic cells, suggest why only about 1% of genetic information is transcribed into functional RNA sequences in most eukaryotic cells. [2]

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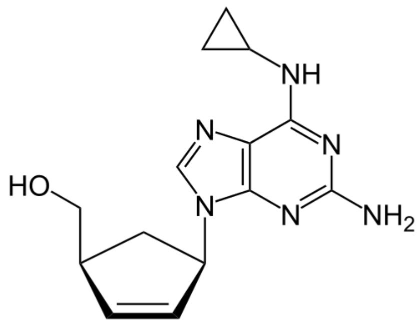
[Total: 11]

4. The HIV virus was first discovered in 1984. It is found to be able to infect human immune cells.

(a) Describe how HIV enters a host cell. [3]

With modern medicine, HIV-positive patients can now live longer life. Highly Active Antiretroviral Therapy (HAART) whereby patients take two or more anti-retrovirus drug concurrently has been proven to be an effective therapy for HIV patients. One of the drugs known to be used in HAART is Abacavir.

Fig 4.1 shows the structure of Abacavir, which is a nucleoside reverse transcriptase inhibitor.



Abacavir

(b) Suggest how Abacavir can prevent HIV. [3]

(c) State two ways in which the life cycle of lambda bacteriophage is different from HIV? [2]

[Total: 8]

5. Mutations in the *BRCA1*, which is a tumour suppressor gene, are associated with an increased risk of breast cancer. Researchers have identified hundreds of mutations in the gene. Women with *BRCA1* gene mutations are associated to an 80% risk of developing breast cancer by age 90 and increased risk of developing ovarian cancer.

BRCA1 gene encode a nuclear protein in human. BRCA1 protein contains two functional domains: an N-terminal Ring domain and a C-terminal BRCT domain. The C-terminal BRCT domain of BRCA1 is a DNA binding domain, which is also required for BRCA1's translocation and accumulation at DNA damage sites for DNA repair.

The three alleles of *BRCA1* gene have the following mutations in C-terminal BRCT domain:

- single-base substitution at nucleotide 1047
- single-base insertion at nucleotide 5382
- single-base deletion at nucleotide 6174

(a) **(i)** Identify and explain which mutation(s) has the most serious consequences for the structure of the BRCA1 protein. [3]

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(ii) Explain how the change in **(a)(i)** could result in cancer. [2]

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The *HER2* gene is another gene that link to breast cancer. HER2 protein is a growth factor receptor on the surface of normal breast cells. When the growth factor binds to HER2 protein, the cell is stimulated to grow and divide. Gene amplification of *HER2* gene is found in cancer cells.

Breast cancer targeted therapy uses drugs that block the growth of breast cancer cells. Trastuzumab is one of the drugs being used in the therapy.

Fig. 5.1 shows the structure of HER2 protein and Trastuzumab.

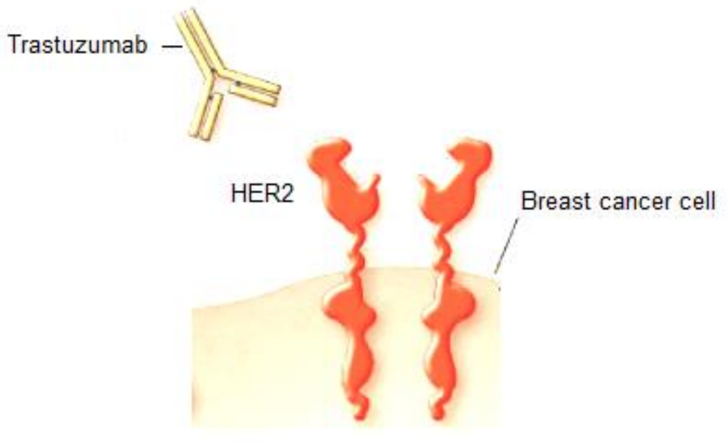

Fig. 5.1

(b) Suggest how amplification of *HER2* gene could result in breast cancer. [2]

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(c) Suggest how Trastuzumab works in cancer treatment. [2]

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[Total: 9]

Part I

6. In an investigation, animal cells were exposed to different concentrations of glucose. The rate of update of glucose into cells across the cell surface membrane was determined for each concentration. Fig 6.1 shows the results.

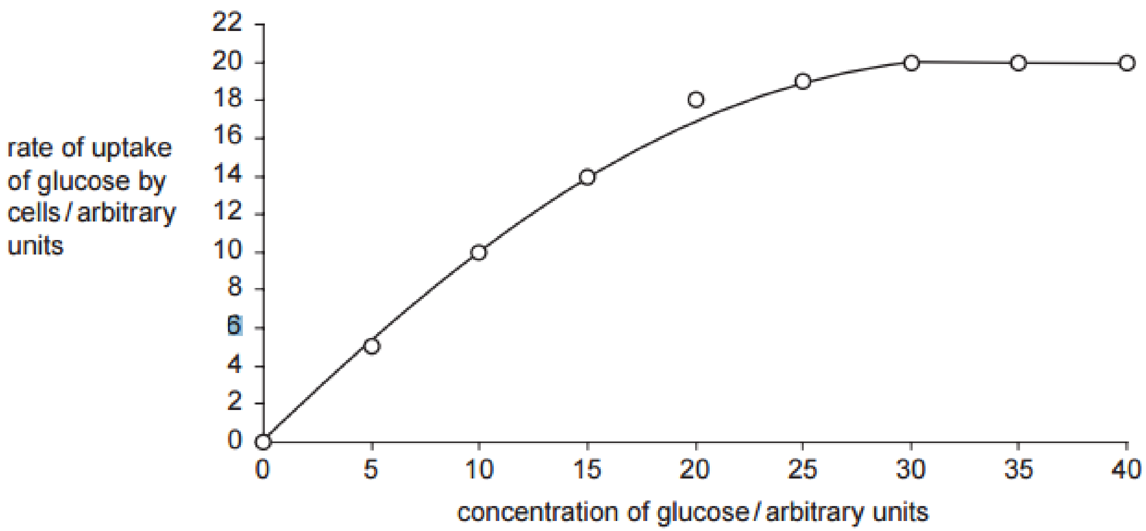

Fig 6.1

(a) Using the information from Fig 6.1,

(i) name the process by which glucose is taken up into the cell. [1]

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(ii) give reasons to support your answer in **(a)(i)**. [2]

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Part II

Following digestion, glucose is transported via the bloodstream and taken up by animal cells such as liver cells. In an investigation, mammalian liver cells were homogenised and the resulting homogenate centrifuged.

Samples of the complete homogenate and samples containing only the mitochondria or only the cytosol were incubated with:

1. Glucose
2. Pyruvate
3. Glucose and cyanide
4. Pyruvate and cyanide

After incubation, the presence or absence of carbon dioxide and lactate in each sample was determined. The results are shown in Fig 6.2.

	samples of homogenate					
	complete		only mitochondria		only cytosol	
	carbon dioxide	lactate	carbon dioxide	lactate	carbon dioxide	lactate
1 glucose	✓	✓	✗	✗	✗	✓
2 pyruvate	✓	✓	✓	✗	✗	✓
3 glucose and cyanide	✗	✓	✗	✗	✗	✓
4 pyruvate and cyanide	✗	✓	✗	✗	✗	✓

✗ = absent ✓ = present

Fig 6.2

(b) Explain why carbon dioxide is produced when the sample with mitochondria only is incubated with pyruvate but not when it is incubated with glucose. [3]

In the presence of cyanide, lactate is produced but carbon dioxide is not. In addition, the oxygen level did not decrease.

(c) Suggest an explanation for the observation. [4]

[Total: 10]

7. Coat colour in horses may be black, chestnut or gray. These colour differences are controlled by two independently assorting genes. In the absence of coloured pigments, the coat of the horse is gray.

A farmer, Zheng, crossed a pure-breeding stallion with chestnut coat and a pure-breeding mare with gray coat.

All the F_1 offspring were horses with gray coat. The offspring were allowed to interbreed and the results of the cross were recorded according to the coat colour of the horses:

Horses with gray coat	130
Horses with black coat	34
Horses with chestnut coat	12

(a) Using the symbols below, draw a genetic diagram to show the cross described. [4]

G – Gray **g** – Coloured **B** – Black **b** – Chestnut

(b) Explain the interaction between the gene loci that leads to the ratio observed in **(a)**. [4]

(c) A chi-squared test was performed on the results of the cross, and the calculated χ^2 value obtained is 0.152.

Table 7.1

Degrees of freedom	probability				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

Using the χ^2 value provided and Table 7.1, describe what conclusion may be drawn. [2]

[Total: 10]

8. The Death Valley region of Nevada in the USA used to have an extensive lake system.

Approximately 20 000 years ago the lakes started to dry up and now consist of isolated small pools. Four different species of the desert pupfish have been found living in these pools.

Evidence indicates that over 20 000 years ago there was only one species of pupfish living in the lake system.

Fig. 8.1 shows the lake 20 000 years ago and a common pupfish found in the lake.

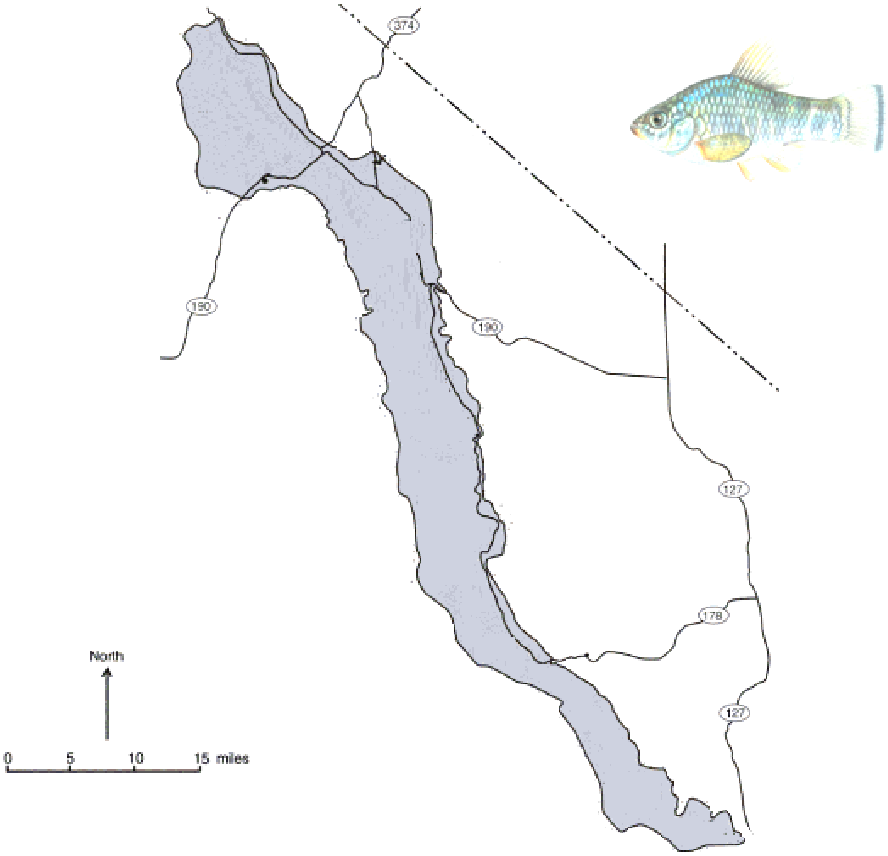

Fig. 8.1

(a) Explain how the change from an extensive lake system to just a few pools could have resulted in the evolution of four new species of desert pupfish. [4]

(b) State how environmental factors can act as stabilising forces of natural selection in an isolated pool, after the initial evolution of a new species of desert pupfish. [2]

Fig. 8.2 shows the nucleotide base sequence of a length of DNA from the gene coding for cytochrome b from the four pupfish species. The nucleotide bases with no change with reference to species 1 are indicated by a dot.

Species	**Nucleotide base**																				
1	**G**	**G**	**A**	**T**	**G**	**A**	**T**	**A**	**T**	**G**	**G**	**G**	**G**	**G**	**A**	**T**	**G**	**T**	**T**	**G**	**C**
2	•	•	•	**G**	•	•	•	•	•	•	•	**A**	•	**A**	•	•	**A**	•	•	•	•
3	**C**	**A**	**G**	**C**	•	•	•	•	•	•	**A**	**A**	**A**	•	•	•	**A**	•	**G**	**A**	**T**
4	•	•	•	**G**	•	•	**G**	•	•	•	•	**A**	•	**A**	•	•	**A**	•	•	•	**T**

Fig. 8.2

(c) **(i)** Suggest why the cytochrome b gene is used to measure changes in DNA sequences in closely related species. [2]

(ii) Explain how the nucleotide sequences in Fig 8.2 can be used to determine the evolutionary relationships between the four pupfish species. [2]

[Total: 10]

Section B (20 marks)

Answer **one** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

A **NIL** return is necessary if you have not attempted this section.

9. **(a)** Compare and contrast collagen and haemoglobin. [7]

(b) Discuss how control elements influence transcription. [7]

(c) Describe mode of action of enzymes, such as DNA polymerase. [6]

[Total: 20]

10. **(a)** Explain the sentence 'The expression of lactose metabolising proteins in *E. coli* is under positive and negative gene regulation'. [8]

(b) Describes the role of insulin in the regulation of blood glucose level in human. [8]

(c) Outline the advantages of signal transduction. [4]

[Total: 20]